Petnica DTW

Preparation plan / August 20-26, 2026

A thin, complete and explainable pipeline before departure - with August 26 protected for review, rehearsal and confidence-building.

Where things stand

~1 day
ahead overall

76
usable Es3 recordings

0
preprocessing failures

HONEST STATUS

Core centring and body-size normalization were implemented about two calendar days early, and validated on every usable Es3 recording. The project is only about one working day ahead overall because subject-wise evaluation, the DTW baseline and the reproducible runner are still not built.

DATE	PRIMARY FOCUS	STATUS
Aug 20	Environment and dataset inventory	Complete
Aug 21	Manifest, loader and early preprocessing	Core complete
Aug 22	Understand preprocessing; exercise choice; grouping	Next
Aug 23	Finish preprocessing decisions and visual checks	Partly done
Aug 24	Plain DTW baseline	Planned
Aug 25	Subject-wise evaluation and reproducible run	Planned
Aug 26	Code revision, walkthrough and rehearsal	Protected

Completed foundation

AUG 20 Environment and dataset inventory COMPLETE

- Created the project structure, Git repository and Python environment.
- Kept the original KIMORE data read-only and mapped its folder structure.
- Inspected subjects, cohorts, exercise folders, skeleton format, sequence lengths, scores and missing material.
- Created an audit script, dataset manifest and summary of exclusions.
- Confirmed that an execution contains a 25-joint Kinect sequence, tracking states, metadata and a clinical TS target when available.

End-of-day output: A defensible inventory of KIMORE and a clear description of one sample.

AUG 21 Manifest, loader and early preprocessing CORE COMPLETE

- Walked through the manifest and loader line by line, including the reshape from (frames, 100) to (frames, 25, 4).
- Loaded XYZ positions as (frames, 25, 3) and tracking states as (frames, 25).
- Checked usable Es3 data: 76 executions; E_ID17_Es3 excluded for a missing target; S_ID5_Es3 excluded for a missing JointPosition file.
- Compared torso length with shoulder width as body-scale candidates. Torso was more stable in 70 of 76 recordings and fully tracked in every usable frame.
- Implemented centring on SpineBase and normalization by median SpineBase-to-SpineShoulder length.
- Validated shape preservation, tracking-state preservation, finite coordinates, zero-valued SpineBase and median normalized torso length of 1.0 across all usable Es3 recordings.

End-of-day output: Working and dataset-wide validated core preprocessing.

Scope note: A basic signal/preprocessing visualization is deliberately carried into August 23 so it can be understood rather than rushed.

EVIDENCE BEHIND THE SCALE CHOICE

Torso median relative MAD: 3.16%; shoulder median relative MAD: 6.46%. Torso usable fraction: 100% in every recording; shoulder usable fraction: median 69.65%, minimum 29.24%. This supports torso length as a stable and explainable baseline, not as a universal claim about human proportions.

Understanding and preprocessing

AUG 22

Relearn preprocessing; lock Es3; guard against leakage

NEXT

- Restart preprocessing from first principles: raw array layout, positions, tracking states, centring, NumPy broadcasting, bone lengths, median, MAD and body-size normalization.
- Explain why centring removes camera-relative translation without changing distances between joints.
- Explain why one 3D torso scale is used for the entire recording and why frame-by-frame body orientation normalization could erase trunk rotation in Es3.
- Document why Es3 is the initial exercise: sufficient usable subjects, available clinical TS targets, workable score variation, reliable recordings and interpretable trunk-rotation signals.
- Create subject groups for evaluation and add a programmatic assertion that no subject can appear in both training and testing.
- Review the diagnostic and preprocessing scripts until every important line can be described in plain language.

End-of-day output: A short Es3-selection note, subject-group utility, leakage test and a personally understood preprocessing pipeline.

Scope note: No DTW implementation today. Understanding and evaluation safety are the priority.

AUG 23

Finish defensible preprocessing and visual verification

PARTLY DONE

- Keep translation relative to SpineBase and median torso-length normalization already implemented.
- Inspect missing or poorly tracked joint values and choose an explicit treatment only if the data show that one is needed.
- Decide whether light smoothing is justified by observed noise; avoid adding it automatically.
- Do not rotate each frame into a common body orientation because Es3 measures trunk rotation. Only consider a fixed camera-alignment correction if camera orientation actually varies.
- Do not force all sequences to the same length for DTW unless an observed problem requires it; DTW is intended to handle duration differences.
- Create before/after plots for at least one representative signal and one skeleton-derived measurement.
- Reserve any learned feature scaling for training folds only, so test data cannot influence preprocessing parameters.
- Run the finalized checks across all 76 usable recordings.

End-of-day output: Final preprocessing decisions, visual evidence and dataset-wide validation.

Scope note: If missing-data handling or smoothing is unnecessary, record that decision instead of inventing extra processing.

PREPROCESSING PRINCIPLE

Every transformation must solve an observed problem, preserve the movement being assessed and be explainable during mentor review. More preprocessing is not automatically better preprocessing.

Thin end-to-end baseline

AUG 24 Plain DTW baseline PLANNED

- Use only training subjects when selecting a reference execution.
- Start with one high-quality training execution as the simplest understandable template.
- Compare every execution with the template and obtain both the DTW alignment path and distance.
- Use a path-normalized cost, such as aligned RMSE, because raw accumulated DTW cost tends to increase with alignment-path length.
- Calibrate the DTW distance to the clinical TS score using a simple regression fitted only on training data.
- Save sample ID, subject ID, actual score, DTW distance, predicted score and fold/reference metadata.
- Inspect at least one good alignment and one poor alignment before trusting aggregate metrics.

End-of-day output: A plain-DTW prediction pipeline that runs on one subject-wise split and produces inspectable per-sample results.

Scope note: Do not add per-feature DTW, learned weights, template ensembles or CNNs yet.

```
aligned_RMSE = sqrt( sum((x_i - y_j)^2 for (i,j) in path) / len(path) )
```

BASELINE SUCCESS CRITERION

Success on August 24 means that the complete route from loaded sequence to predicted clinical score works without subject leakage. Strong accuracy is not required yet; a poor but honest baseline is useful.

Minimal data flow

Load + preprocess	Training-only template	DTW + aligned RMSE	Simple regression	Predicted TS

Evaluation, packaging and revision

AUG 25

Subject-wise evaluation and reproducible experiment

PLANNED

- Use outer GroupKFold with subject ID as the grouping variable; use subject-wise validation inside training only when model choices require it.
- Verify the subject-disjoint condition programmatically for every fold.
- Report MAE and Spearman as the primary metrics; include RMSE and Pearson as secondary metrics.
- Save fold-wise and overall metrics, plus one row of actual and predicted scores per sample.
- Create at least one diagnostic plot, preferably actual versus predicted score, and preserve enough metadata to reproduce it.
- Provide one command that executes the pipeline from manifest to saved results.
- Add essential tests for subject leakage and the central preprocessing invariants.
- Lock dependencies, add a short README, list known limitations and back up code plus results.

End-of-day output: A reproducible plain-DTW experiment with subject-wise predictions, metrics, saved results and a diagnostic plot.

Scope note: This is the busiest day. Documentation polish is optional if it threatens the working experiment; correctness, saved results and leakage safety are not optional.

```
python run_experiment.py --exercise Es3 --method plain_dtw
```

AUG 26

Understand, revise and rehearse

PROTECTED

- Make no planned major implementation changes and add no new method.
- Walk through the complete pipeline in order: manifest, loader, preprocessing, grouping, template selection, DTW, calibration, evaluation and saved outputs.
- For every file, explain its purpose, inputs, outputs and the reason each important function exists.
- Revisit any line, NumPy operation, metric or design choice that cannot yet be explained confidently.
- Practise explaining the main methodological choices: Es3, SpineBase centring, torso scaling, no frame-wise rotation, path-normalized DTW and subject-wise evaluation.
- Run the complete command once from a clean starting point and confirm that expected files and metrics are produced.
- Review one successful example, one poor example and the current known limitations.
- Fix only blocking bugs or misunderstandings, then make a final backup before leaving.

End-of-day output: A pipeline that runs, plus the ability to explain what it does, why it does it and where it may fail.

Scope note: The 26th is a confidence buffer, not a hidden development deadline.

Departure readiness checklist

DATA	Manifest and exclusions are understood; usable Es3 sample count is reproducible.
LOADER	Shapes, columns, tracking states and loader validations can be explained.
PREPROCESSING	Centring and body-size normalization are understood and validated across the dataset.
LEAKAGE	No subject can occur in both training and testing; the assertion passes for every fold.
DTW	Template selection uses training subjects only; distance is normalized by alignment-path length.
CALIBRATION	The mapping from DTW distance to clinical score is fitted using training data only.
EVALUATION	MAE, Spearman, RMSE and Pearson are produced subject-wise and saved.
REPRODUCIBILITY	One command recreates predictions, metrics and at least one plot.
UNDERSTANDING	Every important file and methodological decision can be explained without relying on memorized wording.
BACKUP	Code, dependency list, results, figures, known problems and notes are safely backed up.

DEFINITION OF DONE FOR AUGUST 26

The goal is not a finished research project. The goal is a small, leakage-safe and reproducible baseline that Luka can genuinely understand, defend and extend with his mentor at Petnica.

Technical references

DTW and alignment paths: tslearn DTW documentation
Subject-wise folds: scikit-learn GroupKFold documentation